

Week 1 - JUnit_Basic Testing Exercises

Exercise 1: Setting Up JUnit Scenario:

You need to set up JUnit in your Java project to start writing unit tests. Steps: 1. Create a new Java project in your IDE (e.g., IntelliJ IDEA, Eclipse). 2. Add JUnit dependency to your project. If you are using Maven, add the following to your pom.xml: junit junit 4.13.2 test 3. Create a new test class in your project.

Calculator.java

```
public class Calculator {  
    public int add(int a, int b) {  
        return a + b;  
    }  
  
    public int subtract(int a, int b) {  
        return a - b;  
    }  
  
    public int multiply(int a, int b) {  
        return a * b;  
    }  
  
    public double divide(int a, int b) {  
        if (b == 0) {  
            throw new IllegalArgumentException("Division by zero is not allowed.");  
        }  
        return (double) a / b;  
    }  
}
```

```
CalculatorTest.java import static
```

```
org.junit.Assert.assertEquals; import
```

```
org.junit.Test;
```

```
public class CalculatorTest {
```

```
    @Test    public void testAdd() {
```

```
        Calculator calc = new Calculator();
```

```
        assertEquals(5, calc.add(2, 3));
```

```
    }
```

```
    @Test    public void testSubtract() {
```

```
        Calculator calc = new Calculator();
```

```
        assertEquals(1, calc.subtract(3, 2));
```

```
    }
```

```
    @Test    public void testMultiply() {
```

```
        Calculator calc = new Calculator();
```

```
        assertEquals(6, calc.multiply(2, 3));
```

```
    }
```

```
    @Test(expected = IllegalArgumentException.class)
```

```
    public void testDivideByZero() {    Calculator calc
```

```
= new Calculator();    calc.divide(5, 0);
```

```
    }
```

```
    @Test    public void testDivide() {
```

```
        Calculator calc = new Calculator();
```

```
        assertEquals(2.0, calc.divide(6, 3), 0.001);
```

```
    }
```

```
}
```

Pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.example</groupId>
  <artifactId>demo</artifactId>
  <version>1.0</version>

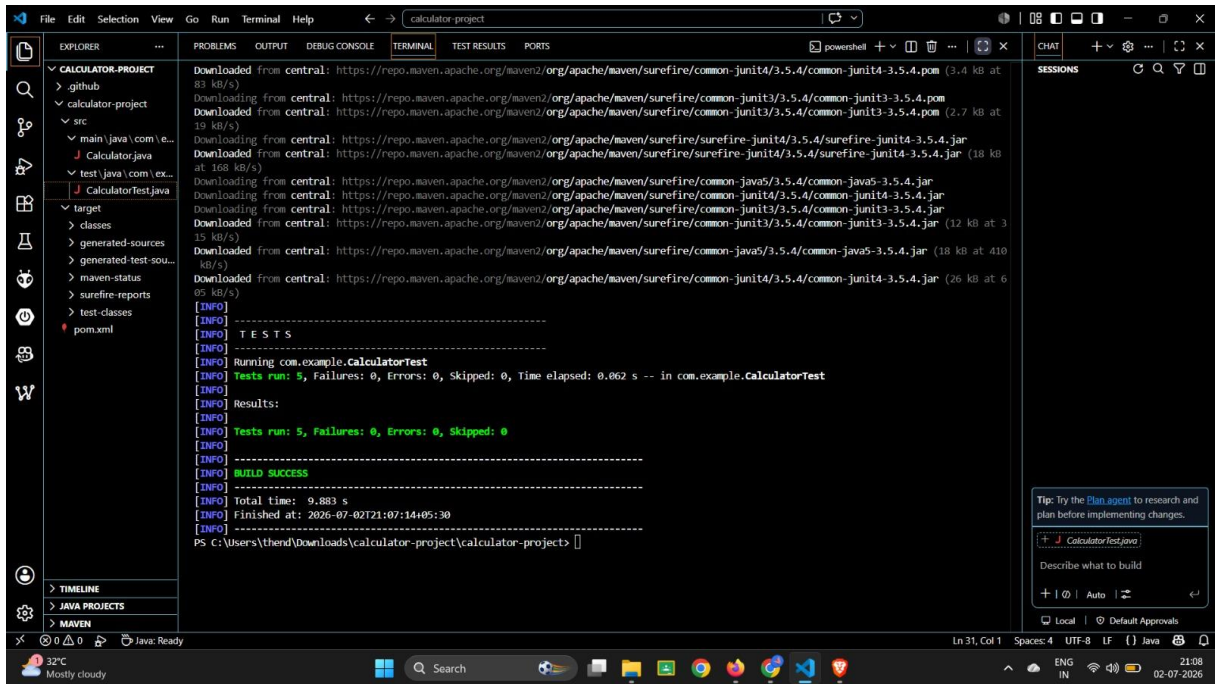
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>4.13.2</version>
      <scope>test</scope>
    </dependency>
  </dependencies>

  <build>
    <plugins>
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-compiler-plugin</artifactId>      <version>3.8.1</version>
        <configuration>
          <source>17</source>
          <target>17</target>
        </configuration>
      </plugin>
    </plugins>
```

</build>

</project>

Output:



The screenshot shows an IDE window with the terminal output of a Maven build. The Explorer on the left shows the project structure: calculator-project, src, main, test, target, classes, generated-sources, generated-test-sources, maven-status, surefire-reports, test-classes, and pom.xml. The terminal output shows the following:

```
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit4-3.5.4/common-junit4-3.5.4.pom (3.4 kB at 83 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit3-3.5.4/common-junit3-3.5.4.pom (2.7 kB at 19 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-junit4/3.5.4/surefire-junit4-3.5.4.jar (18 kB at 168 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-java5/3.5.4/common-java5-3.5.4.jar (12 kB at 315 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit4/3.5.4/common-junit4-3.5.4.jar (12 kB at 315 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-java5/3.5.4/common-java5-3.5.4.jar (18 kB at 410 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit4/3.5.4/common-junit4-3.5.4.jar (26 kB at 605 kB/s)
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.calculatorTest
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.062 s -- in com.example.calculatorTest
[INFO] Results:
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 9.883 s
[INFO] Finished at: 2026-07-02T21:07:14+05:30
[INFO] -----
Ps c:\Users\thend\Downloads\calculator-project\calculator-project>
```

The status bar at the bottom indicates the current file is 'Ln 31, Col 1' in a Java file, with 4 spaces, UTF-8 encoding, and LF line endings. The system tray shows the temperature is 32°C and the date is 02-07-2026.

Exercise 3: Assertions in JUnit

Scenario:

You need to use different assertions in JUnit to validate your test results.

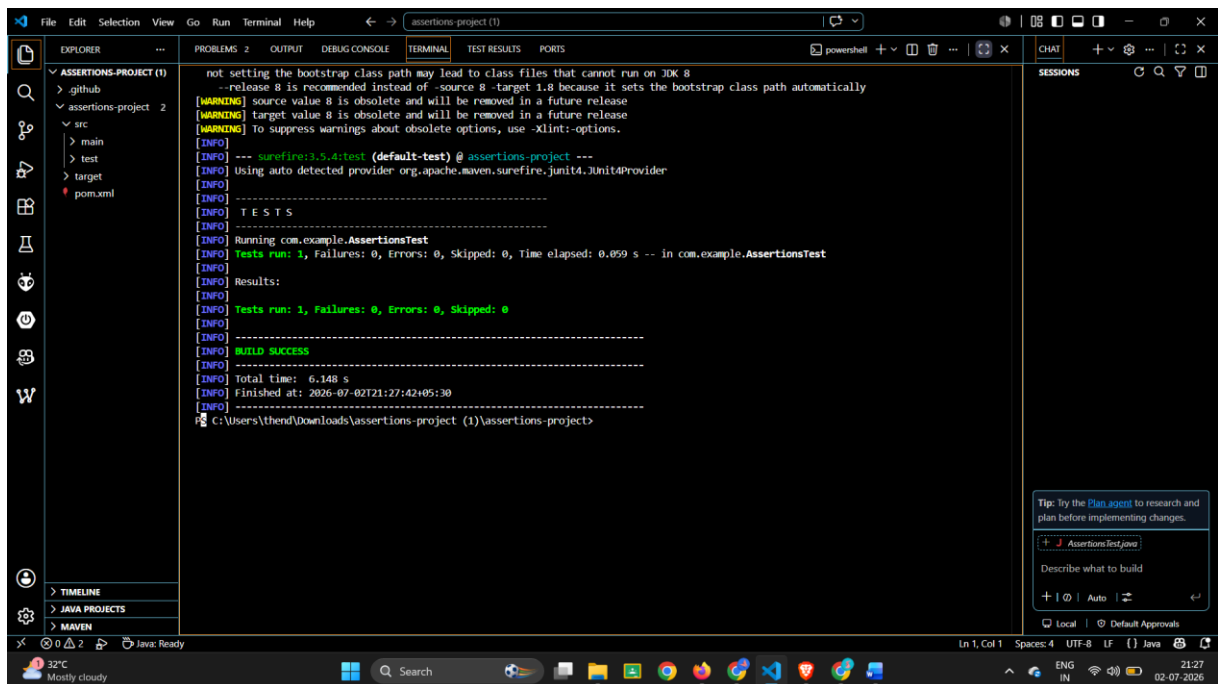
Steps:

1. Write tests using various JUnit assertions.

Solution Code:

```
public class AssertionsTest {  
    @Test    public void  
    testAssertions() {    //  
        Assert equals  
        assertEquals(5, 2 + 3);  
  
        // Assert true  
        assertTrue(5 > 3);  
  
        // Assert false  
        assertFalse(5 < 3);  
  
        // Assert null  
        assertNull(null);  
  
        // Assert not null  
        assertNotNull(new Object());  
    }  
}
```

Output:



The screenshot shows an IDE window titled 'assertions-project (1)'. The left sidebar contains the Explorer, Search, and Run and Debug views. The main editor area displays the output of a Maven test run. The output shows a successful build with the following details:

```
not setting the bootstrap class path may lead to class files that cannot run on JDK 8
--release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automatically
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO] --- surefire:3.5.4:test (default-test) @ assertions-project ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AssertionsTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.059 s -- in com.example.AssertionsTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 6.148 s
[INFO] Finished at: 2026-07-02T21:27:42+05:30
[INFO] -----
C:\Users\thend\Downloads\assertions-project (1)\assertions-project>
```

The right sidebar shows the Chat view with a tip to try the [Plan agent](#) to research and plan before implementing changes. Below the tip is a search bar with the text 'Describe what to build' and a button labeled 'Auto'.

The bottom status bar shows the current file is 'Ln 1, Col 1' with 'Spaces: 4', 'UTF-8', 'LF', and 'Java' encoding. The system tray at the bottom shows the temperature as 32°C, weather as 'Mostly cloudy', and the date and time as 21:27 on 02-07-2026.

Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in JUnit

Scenario:

You need to organize your tests using the Arrange-Act-Assert (AAA) pattern and use setup and teardown methods.

Steps:

1. Write tests using the AAA pattern.
2. Use `@Before` and `@After` annotations for setup and teardown methods.

Calculator.java

```
public class Calculator {  
    public int add(int a, int b) {  
        return a + b;  
    }  
  
    public int subtract(int a, int b) {  
        return a - b;  
    }  
  
    public int multiply(int a, int b) {  
        return a * b;  
    }  
  
    public double divide(int a, int b) {  
        if (b == 0) {  
            throw new IllegalArgumentException("Division by zero is not allowed.");  
        }  
        return (double) a / b;  
    }  
}
```

```
}
```

```
CalculatorTest.java import org.junit.After;  
import static org.junit.Assert.assertEquals;  
import org.junit.Before; import  
org.junit.Test;
```

```
public class CalculatorTest {
```

```
    private Calculator calc;
```

```
    @Before    public void setUp() {        calc =  
new Calculator(); // Arrange (shared)  
    }
```

```
    @After    public void  
tearDown() {        calc =  
null; // Cleanup  
    }
```

```
    @Test    public void  
testAddition() {  
        // Act        int result =  
calc.add(4, 6);  
  
        // Assert  
assertEquals(10, result);  
    }
```



```

@Test    public void
testSubtraction() {

    // Act

    int result = calc.subtract(9, 5);


    // Assert

    assertEquals(4, result);

}
}

```

Output:

The screenshot shows an IDE window with the following content:

- EXPLORER:** Shows a project structure with folders like `github`, `exercise4-aaa-pr...`, `src`, `main`, `test\java\com\ex...`, `CalculatorTest.java`, `target`, and `pom.xml`.
- TERMINAL:** Displays the output of a Maven test run. It includes warnings about obsolete options and source/target values, followed by a successful test execution.


```

[WARNING] To suppress warnings about obsolete options, use -Xlint:options.
[INFO] --- resources:3.4.0:testResources (default-testResources) @ exercise4-aaa-project ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\thend\Downloads\aaa-project\exercise4-aaa-project\src\test\resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ exercise4-aaa-project ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
[WARNING] not setting the bootstrap class path may lead to class files that cannot run on JDK 8
--release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automatically
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:options.
[INFO] --- surefire:3.5.4:test (default-test) @ exercise4-aaa-project ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running com.example.CalculatorTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.065 s -- in com.example.CalculatorTest
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 2.163 s
[INFO] Finished at: 2026-07-02T21:41:31+05:30
[INFO]

```
- STATUS BAR:** Shows "Ln 1, Col 1", "Spaces: 4", "UTF-8", "LF", "Java", and "21:42 02-07-2026".